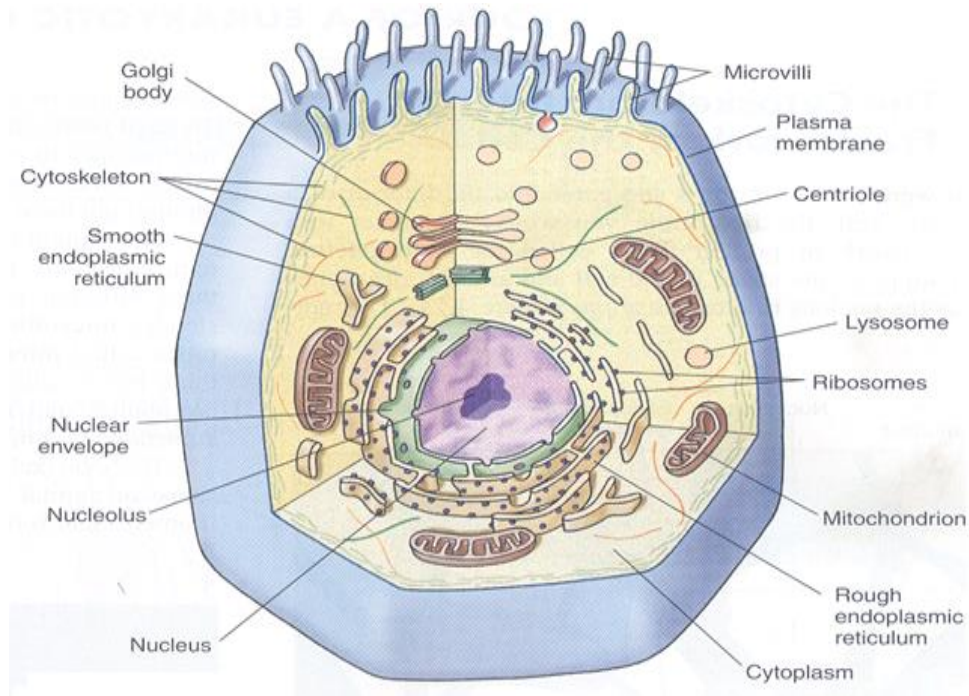


Introduction to Human Physiology



Cell Physiology

Contents

- Cell structure and composition
 - Function of cellular organelles
- Structure and function of cell membrane:
 - Components of plasma membrane
 - Membrane transport of molecules
 - Diffusion
 - Active transport + Bulk transport
- Cellular environment: Body fluids and electrolyte balance
- Physiological adjustment mechanisms of the body

The Cell

- Composition:
 - **CHOs** $\rightarrow \approx 3\%$ of the dry mass of a typical cell
 - **Lipids** $\rightarrow \approx 40\%$ of the dry mass of a typical cell
 - **Proteins** $\rightarrow \approx 50\text{-}60\%$ of the dry mass of a typical cell
 - **Nucleic acids** $\rightarrow \text{DNA} + \text{RNA}$

The Cell

- **Functions:**
 - Obtaining food and O_2 from its surrounding
 - Elimination of waste products
 - Performing various chemical reactions
 - Synthesis of materials for cellular structure, growth and cellular functions
 - Sensitivity and responsiveness to changes in its immediate surrounding

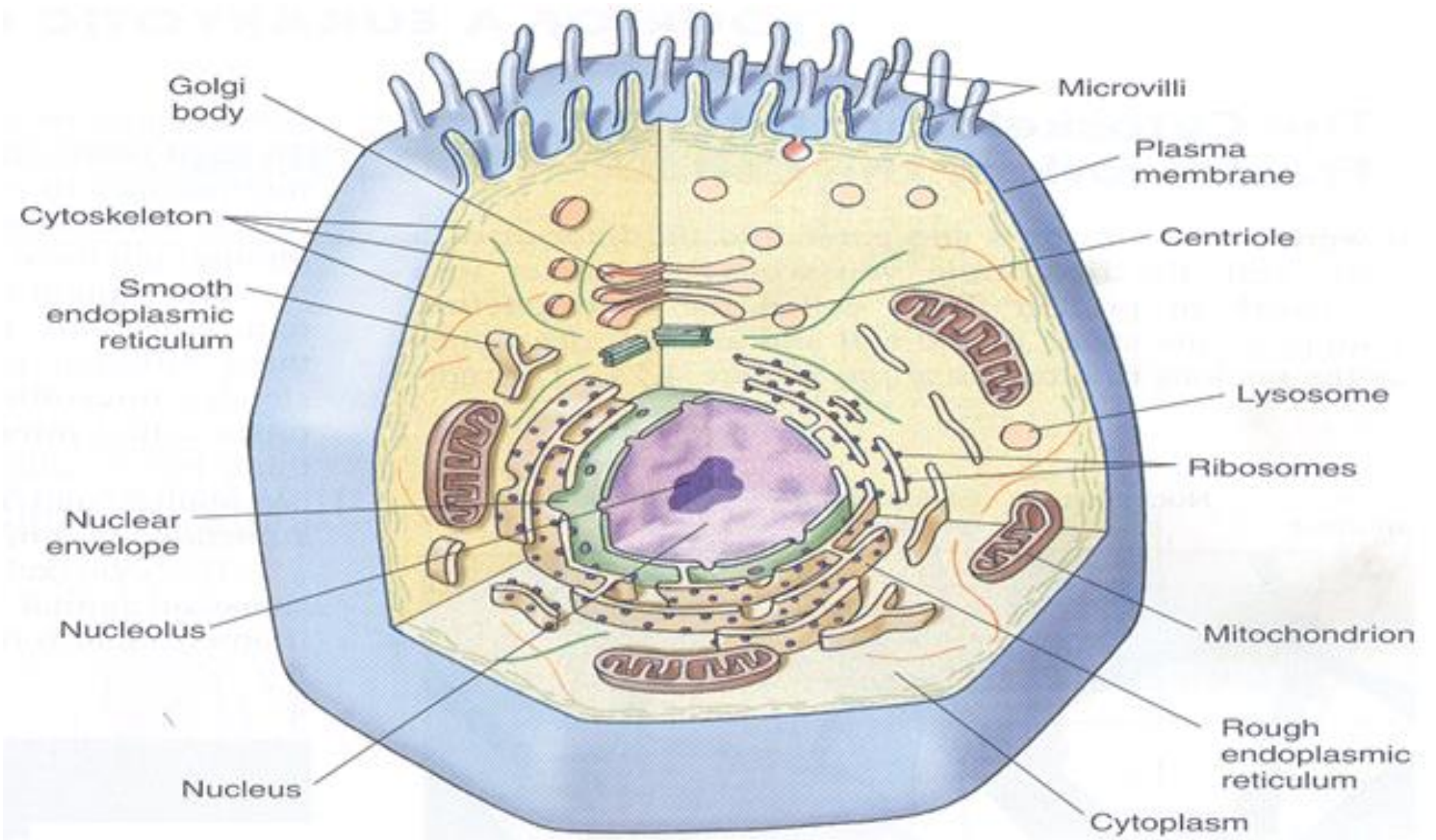
The Cell

- **Functions:**
 - Reproducing (most cells except nerve and muscle cells)
 - Moving materials in and out of the cell and within the cell in carrying out cellular activities
 - The cell membrane serves as the site of **signal transduction** → a process by which a cell converts an input signal into a response

Cell Organelles

- Non-membrane limited:
 - Chromosomes
 - Nucleoli
 - Ribosomes
 - Microtubules
 - Microfilaments
 - Centrioles
- Membrane limited organelles:
 - Nucleus
 - Endoplasmic reticulum (ER)
 - Golgi apparatus (GA)
 - Lysosomes
 - Mitochondria
 - Peroxisomes

Cell Organelles



Cell Organelles

- Plasma Membrane
 - 7.5nm thick, Composed of a phospholipids bilayer and protein
 - Selective barrier to movement of ions/molecules into & out of cell
- Mitochondria
 - Rod-or oval-shaped bodies surrounded by two membranes.
 - Major site of ATP production, oxygen utilization, and CO₂ formation.
 - Contains enzymes of Krebs cycle and oxidative phosphorylation

Cell Organelles

- Endoplasmic Reticulum
 - 1 interconnected cell organelle
 - Two opposed membranes separating a space
- Granular ER
 - Ribosomal particles bound to ER membrane
 - Synthesis of proteins to be secreted from cell
- Agranular ER
 - No ribosomes
 - Fatty acid and steroid synthesis
 - Calcium storage and release in muscle cells

Cell Organelles

- Golgi apparatus
 - Cup-shaped series of closely opposed membranous sacs and vesicles
 - Concentration and modification of proteins prior to secretion
- Secretory vesicles
 - Membrane-bound sacs containing concentrated solution of proteins
 - Protein secretion

Cell Organelles

- **Lysosomes**
 - Density staining oval body surrounded by membrane
 - Contains hydrolytic enzymes
 - Digestive organelle
 - Specialized for breakdown of engulfed bacteria and damaged cell organelles
- **Peroxisomes**
 - Play important roles in metabolism, ROS detoxification

Cell Organelles

- Free Ribosomes
 - 20-nm-diameter particles composed of RNA and protein. Not bound to membranes.
 - Site at which amino acids are assembled into proteins.
 - Site at which proteins to be used intracellular are assembled

Cell Organelles

- **Bound Ribosomes/filaments**
 - Bound to membranes of granular endoplasmic reticulum.
 - 5-15nm diameter protein threads of variable length
 - Site at which proteins to be secreted from cell are assembled.
 - Cell movements.
 - Also used to provide structural support at cell junctions

Cell Organelles

- **Microtubules**

- 25nm diameter protein tubules with 15nm diameter hollow core
- Maintenance of cell shape (cytoskeleton)
- Associated with movements of cilia, flagella, and mitotic spindle

- **Centrioles**

- Two small cylindrical bodies composed of nine sets of three fused microtubules
- Formation of spindle apparatus at poles of cell during cell division.
- Also associated with formation and movement of cilia

Cell Organelles

- Granules
 - Aggregated of crystals of chemical substances
 - Storage of specialized end products of metabolism
 - Glycogen granules are most common
- Fat droplets
 - Spherical globule of triacylglycerol
 - Storage of fat

Cell Organelles

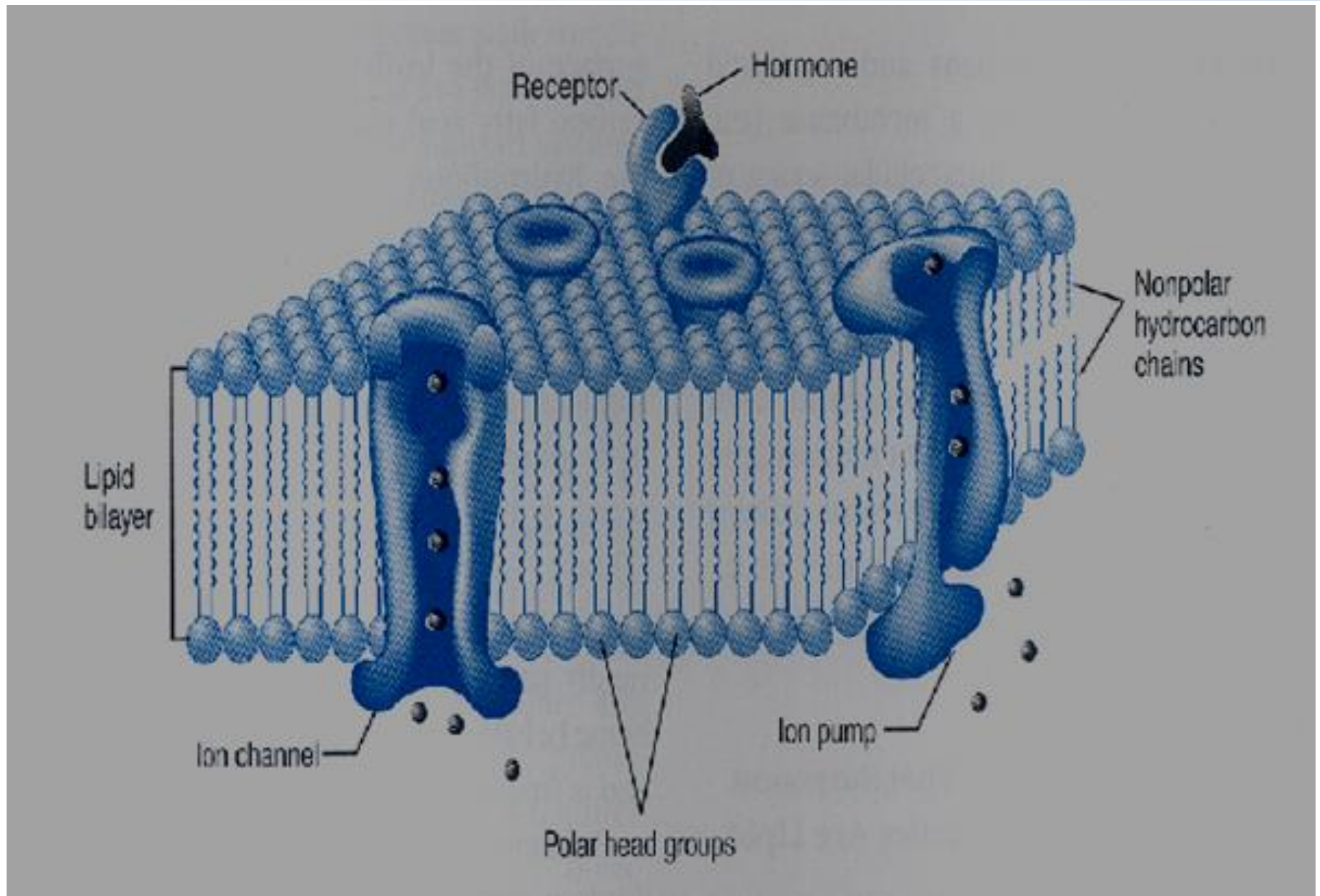
- Nuclear envelope
 - Surrounds the nucleus
 - Two opposed membranes separating small space
 - Nuclear pores: 50-70 nm diameter
 - Barrier to movement of most molecules.
 - Messenger RNA passes to cytoplasm through pores

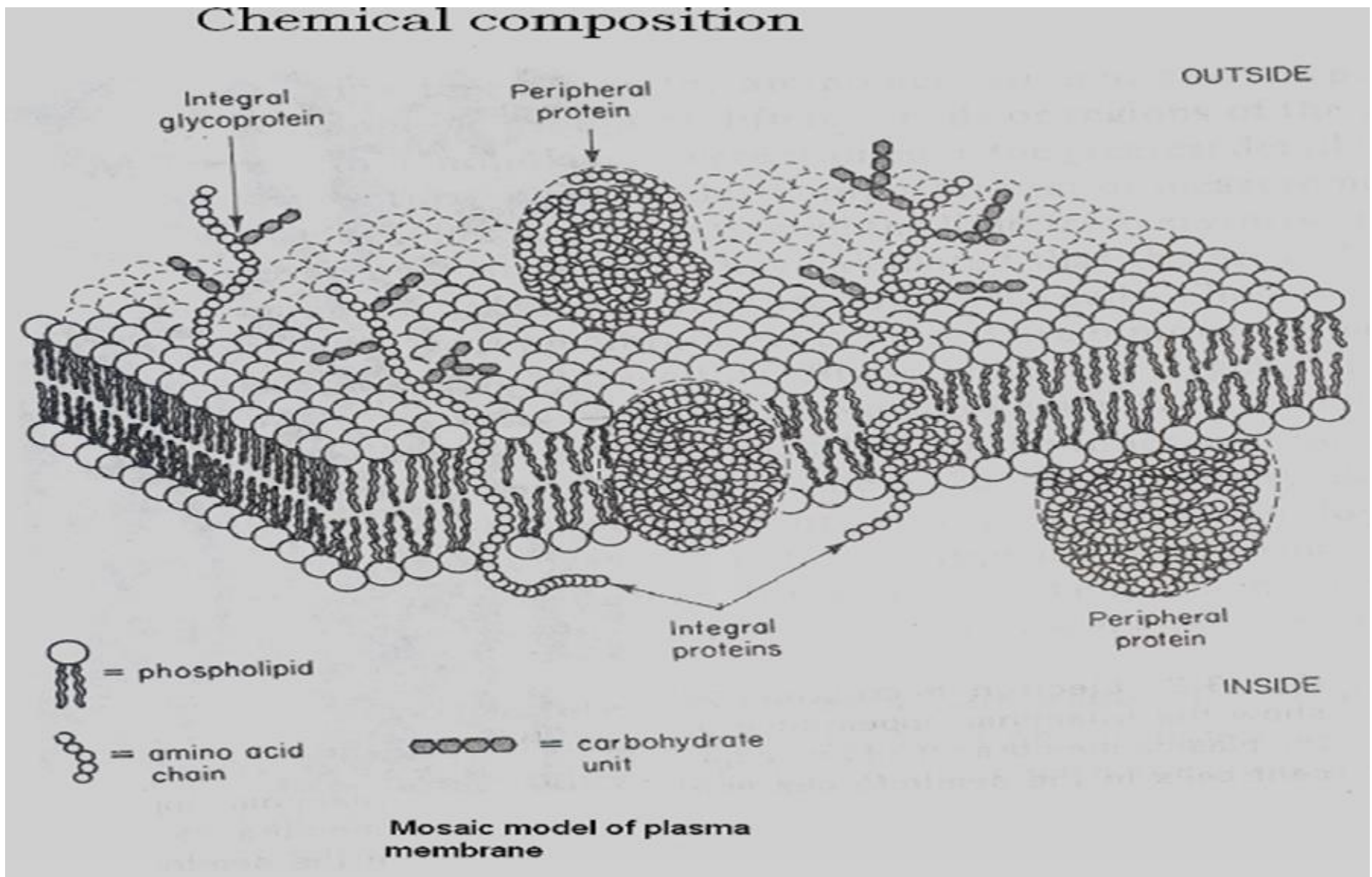
Cell Organelles

- **Chromatin**
 - Coiled threads composed of DNA and protein
 - 46 strands per human cell nucleus
 - DNA stores genetic information.
 - Chromatin condenses into chromosomes during cell division
- **Nucleolus**
 - Coiled filamentous structure associated with granules
 - Not surrounded by membrane
 - Site of ribosomal RNA synthesis

Cell Membrane

- **Composition:**
 1. Lipids → Phospholipids + Cholesterol
 2. Proteins → Peripheral + Integral
 3. CHOs → Polysaccharides





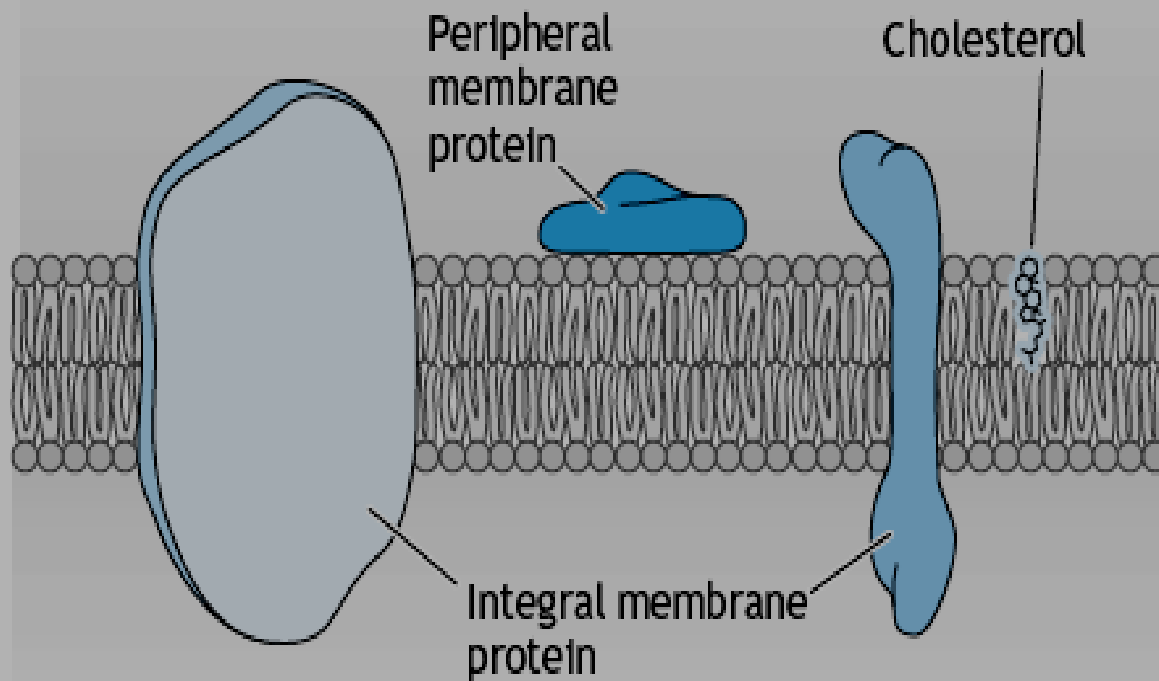


Fig.. Membrane proteins.

Cell Membrane

1. Lipids

a. Phospholipids → *Amphiphilic/amphipathic*

- *Constitute the majority of membrane lipids*
- Have a glycerol backbone (the hydrophilic head) + two fatty acid tails (hydrophobic)
- Hydrophobic tails face each other and form a bi-layer
- Lipid-soluble substances (O₂, CO, steroid hormones...) can dissolve in the hydrophobic lipid bi-layer and cross cell membranes easily

Cell Membrane

a. Phospholipids → *Amphiphilic/amphipathic*

- Water-soluble substances (Na^+ , Cl^- , glucose, H_2O ...) cannot dissolve in the lipid of the membrane, but may cross through water-filled channels/pores, or may be transported by carriers

b. Cholesterol

- Long rigid hydrophobic chain and a small polar hydroxyl group
- Maintain the integrity + fluidity of the cell membrane
- Helps to prevent the membrane from becoming too fluid + too firm

Cell Membrane

b. Cholesterol

- Long rigid **hydrophobic chain** and a small **polar hydroxyl group**
- Interacts with the fatty acid tails and functions to immobilise the outer surface of the membrane (**reducing fluidity**)
- It functions to separate phospholipid tails and so **prevent crystallisation** of the membrane
- It makes the membrane less permeable to very small water-soluble molecules that would otherwise freely cross
- It helps secure peripheral proteins by forming high density lipid rafts capable of anchoring the protein

Cell Membrane

2. Proteins:

a. Peripheral (extrinsic) proteins

- **Hydrophilic** and readily dissociated from the membrane
 - *Not* imbedded and *not* covalently bound to the cell membrane
- Loosely attached to the cell membrane by electrostatic interactions:
 - Free, floating on the surface (stud the inside and the outside of the membrane)

Cell Membrane

a. Peripheral (extrinsic) proteins

- Account for about 30% of the membrane proteins
- Bind specific hormones and proteins on the cell membrane
- Also function as **enzymes**

Cell Membrane

b. Integral (intrinsic) proteins

- Tightly associated with membrane → anchored to and imbedded in the cell membrane through hydrophobic interactions
- Partly **hydrophilic** (polar and protruding to cell surface) and partly **hydrophobic** (non-polar and embedded in the lipid bilayer)
- Exist as separate globular units running through the width of the cell membrane
- Protruding part may often carry CHO chains or lipids attached to their tips

Cell Membrane

b. Integral (intrinsic) proteins

- Account for about 70% of the membrane proteins
- Serve as:
 - Receptors + Ion channels
 - Transport proteins (carriers)
 - Enzymes + GTP-binding proteins (G proteins)
- Many of them provide **structural channels/pores** through which water molecules and water-soluble substances, especially ions can diffuse in and out of the cell

Cell Membrane

3. Carbohydrates

- Hydrophilic polysaccharides covalently linked to proteins (glycoproteins) or lipids (glycolipids)
- Function as cell recognition markers and adhesion
 - Cell-cell signaling or
 - Cell-pathogen interactions
- Structural role as a physical barrier

Intercellular Connections

a. Tight junctions (*zonula occludens*)

- Intercellular adhesion complexes that control paracellular permeability (size and charge-selective)
 - Tight (impermeable) → renal distal tubule
 - Leaky (permeable) → renal proximal tubule and gallbladder
- Form the border between the **apical and basolateral cell surface** domains in polarized epithelia → **maintenance of cell polarity** by restricting intermixing of the two domain components

Intercellular Connections

b. Gap junctions

- Clusters of intercellular channels that allow direct diffusion of ions and small molecules between **adjacent cells**
- Formed by head-to-head docking of hexameric assemblies (connexons) of tetraspan integral membrane proteins (connexins)
- Formed by two connecting trans-membrane protein rings (**connexins**)
- Permit changes in **membrane potential** (action potential) to pass from cell to cell

Intercellular Connections

c. Desmosomes

- Intercellular junctions that tether intermediate filaments to the plasma membrane
- Critical for maintaining stable cell–cell adhesion
- Found in tissue that experience intense mechanical stress, such as cardiac muscle, bladder, gastrointestinal mucosa, and epithelia

Intercellular Connections

d. Adherence junctions (zonula adherens)

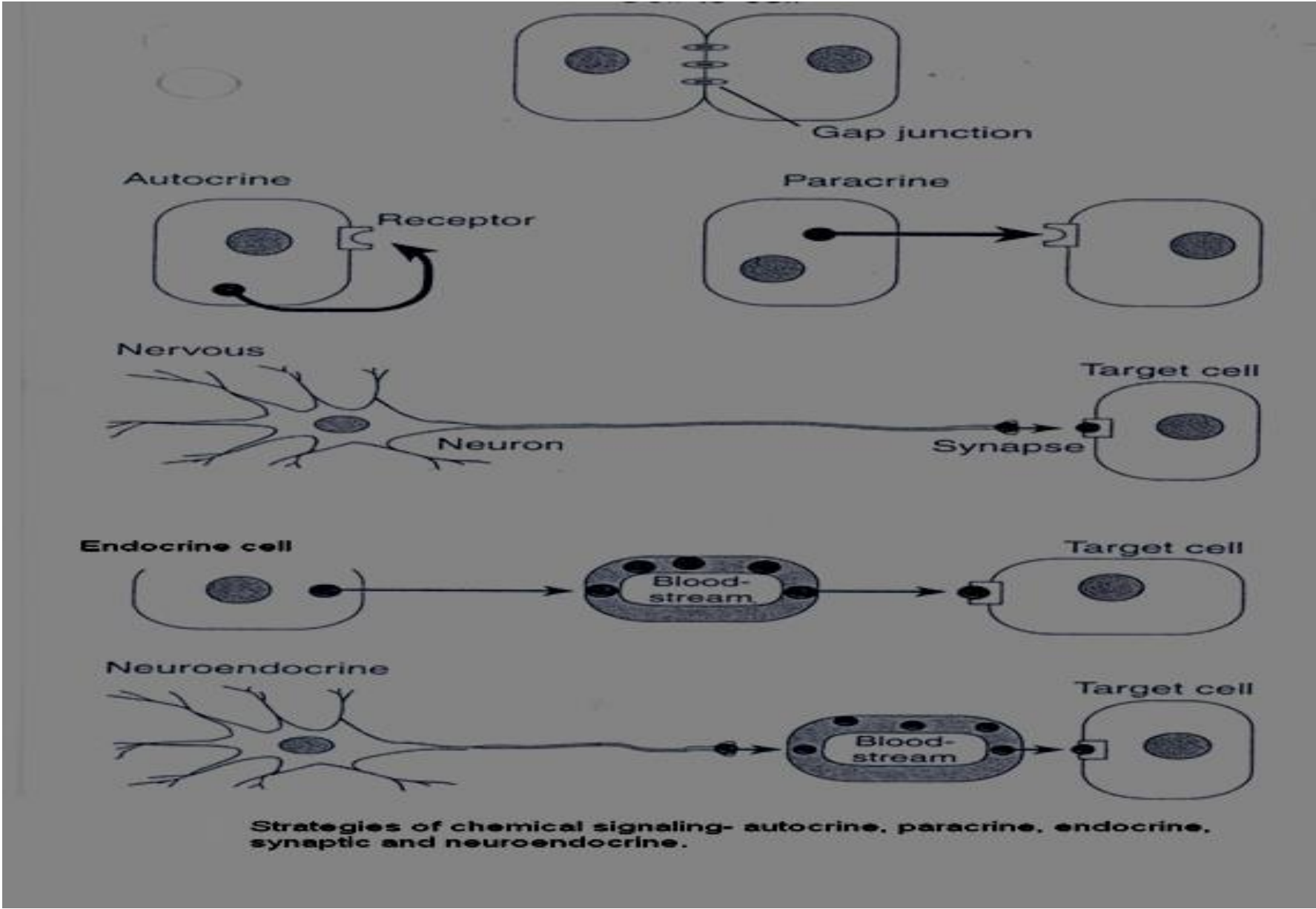
- Cell-cell adhesion complexes that are continuously assembled and disassembled, allowing cells within a tissue to respond to forces, biochemical signals and structural changes in their microenvironment
- Provide strong **mechanical attachments** b/n adjacent cells like cardiac muscle cells

Receptors

- Proteins located in the outer membrane of the cell or within the cell
- **Cytoplasmic & nucleus receptors** → for lipid-soluble messengers
- **Cell surface receptors** → for lipid-insoluble messengers
 - **Channel-linked** receptors (neurotransmitter-gated channels)
 - **Catalytic** receptors (mainly tyrosine-specific protein kinases)
 - **G-protein-linked** receptors (plasma membrane-bound enzymes or ion channels)
- **NB:** each of them transduces extra cellular signals in a different way

Cellular Communication

- Almost all cells sense their environment primarily by detecting **chemical or electrical signals** probably by employing surface proteins (receptors) embedded within the plasma membrane
- Mechanisms of chemical signaling:
 - Autocrine
 - Paracrine
 - Synaptic
 - Endocrine and Neuroendocrine



Transport across the Cell Membrane

Transport Mechanisms

- Types of transport mechanisms:
 1. Passive transport
 2. Active transport
 3. Bulk transport

Transport Mechanisms

1. Passive transport

- Involves physiochemical forces that do not require metabolic energy
- Simple and facilitated diffusion

a. Simple Diffusion

- Free movement of a substance from one part of a solution to another occurring in both directions
- Net movement occurs from an area of high concentration to another area of low concentration

Transport Mechanisms

- Factors affecting rate of diffusion
 - Viscosity of solution
 - Mol. wt → light molecules diffuse faster
 - Lipid solubility → lipid soluble molecules can easily cross the cell membranes
 - Electrical charge and electrical gradient
 - Charged molecules move faster in the direction of the opposite charge and slower or not at all in the direction of the same charge
 - Unionized form of molecules with low concentration on opposite side of the membrane diffuse rapidly in that direction

Transport Mechanisms

b. Facilitated diffusion

- Occurs down an electrochemical gradient ("downhill"), similar to simple diffusion
- Does not require metabolic energy (passive)
- More rapid than simple diffusion (carrier-mediated)
 - Exhibits – **stereo-specificity, saturation and competition**

Transport Mechanisms

b. Facilitated diffusion

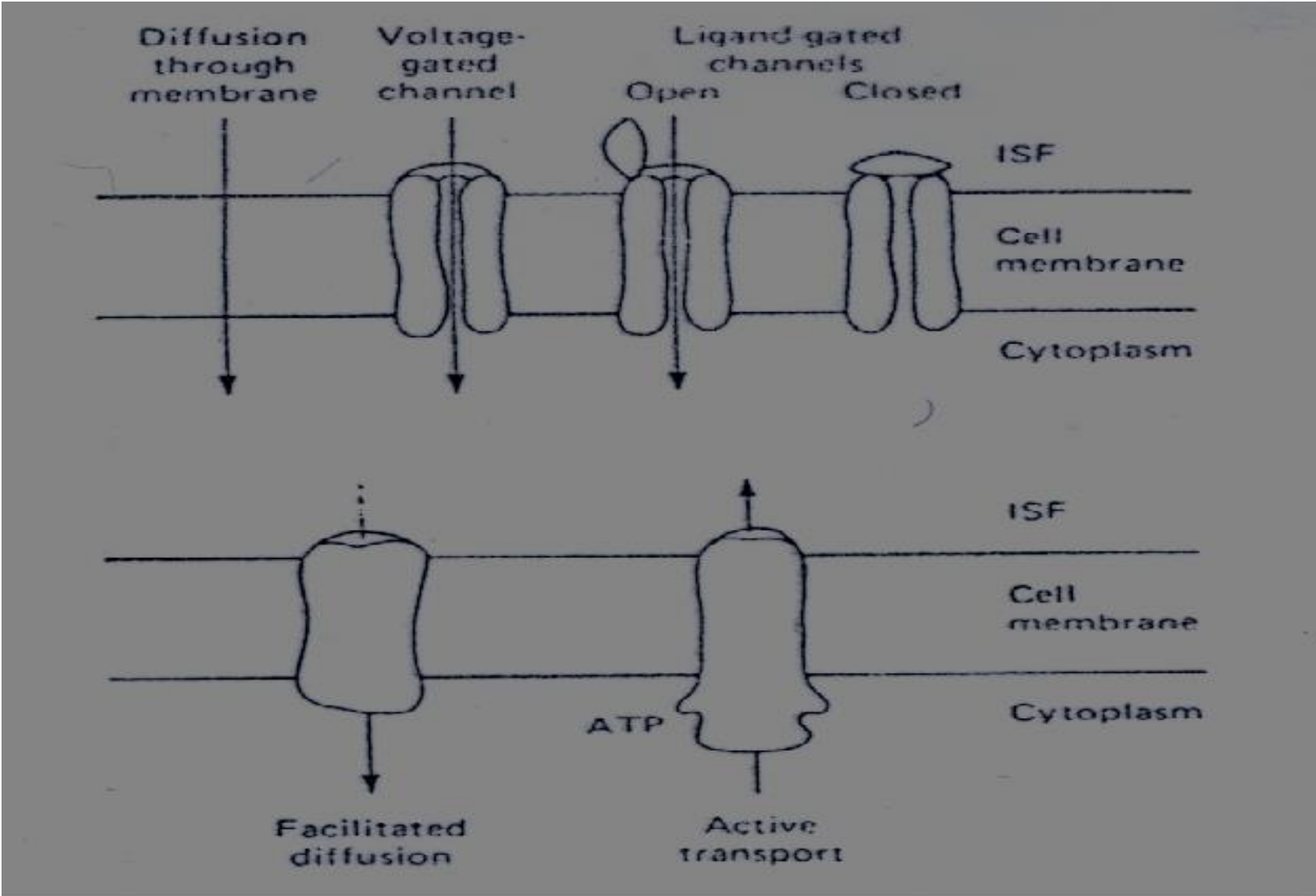
- **Specificity**
 - Membrane reacts with only a selective group of substances
 - Amino acid transporting mechanisms does not transport sugars and vice versa
- **Saturation**
 - Transport increases with concentration only up to a certain point
- **Competition** → occurs between similar molecules that use the same transporter

Transport Mechanisms

- Carrier transport
 - Depending on the number of substances that carriers transport, carrier proteins may be referred to as:
 - **Uniport** → carrying only one substance
 - **Symport** → carrying more than one substance in the same direction
 - **Antiport** → facilitating exchange of substances

Transport Mechanisms

- Ion channels:
 - Leak channels → continuously open
 - Voltage gated
 - Ligand gated:
 - External ligand → hormones, neurotransmitters...
 - Internal ligand → Ca^{++} , cAMP, G-proteins...
 - Mechanically gated → mechanical force



Transport Mechanisms

- **Osmosis:**
 - Passive large water fluxes across a membrane
 - Conditions required:
 - **High conc. of solute** on one side
 - **Lower diffusional permeability** of solute across the membrane
 - **Filtration** → fluid is forced through a membrane because of pressure difference on the two sides
 - A cell kept in:
 - Hypotonic solution ...→ swells
 - Hypertonic solution...→ shrinks

Transport Mechanisms

- Solvent drag
 - A rapid movement of water across membrane in one direction (bulk flow) producing a large movement of solutes
 - Solvent tends to drag along some molecules of solute
 - Minor process found mainly in kidneys

Transport Mechanisms

2. Active transport

- Utilizes metabolic energy
- Includes:
 - Primary active transport
 - a. $3\text{Na}^+-2\text{K}^+$ ATPase (Na^+-K^+ pump)
 - b. Ca^{2+} ATPase (Ca^{2+} pump)
 - c. H^+-K^+ ATPase (proton pump)...
 - Secondary active transport
 - Na^+ -Glucose co-transport
 - $\text{Na}^+-\text{K}^+-2\text{Cl}^-...$

Transport Mechanisms

a. $\text{Na}^+\text{-K}^+$ ATPase ($\text{Na}^+\text{-K}^+$ pump)

- Transports 3Na^+ from ICF to ECF and 2K^+ from ECF to ICF
 - Maintains low intracellular $[\text{Na}^+]$ and high intracellular $[\text{K}^+]$
- Both Na^+ and K^+ are transported against their electrochemical gradients
- Energy is provided from the terminal phosphate bond of ATP.
- **NB**
 - Cardiac glycoside drugs (ouabain + digitalis) specifically inhibit $\text{Na}^+\text{-K}^+$ ATPase

Transport Mechanisms

b. Ca^{2+} ATPase (Ca^{2+} pump)

- Located in the sarcoplasmic reticulum (SR) + the cell membrane
- Transports Ca^{2+} from the ICF to the SR + ECF (against an electrochemical gradient)

c. H^{+} - K^{+} ATPase (proton pump)

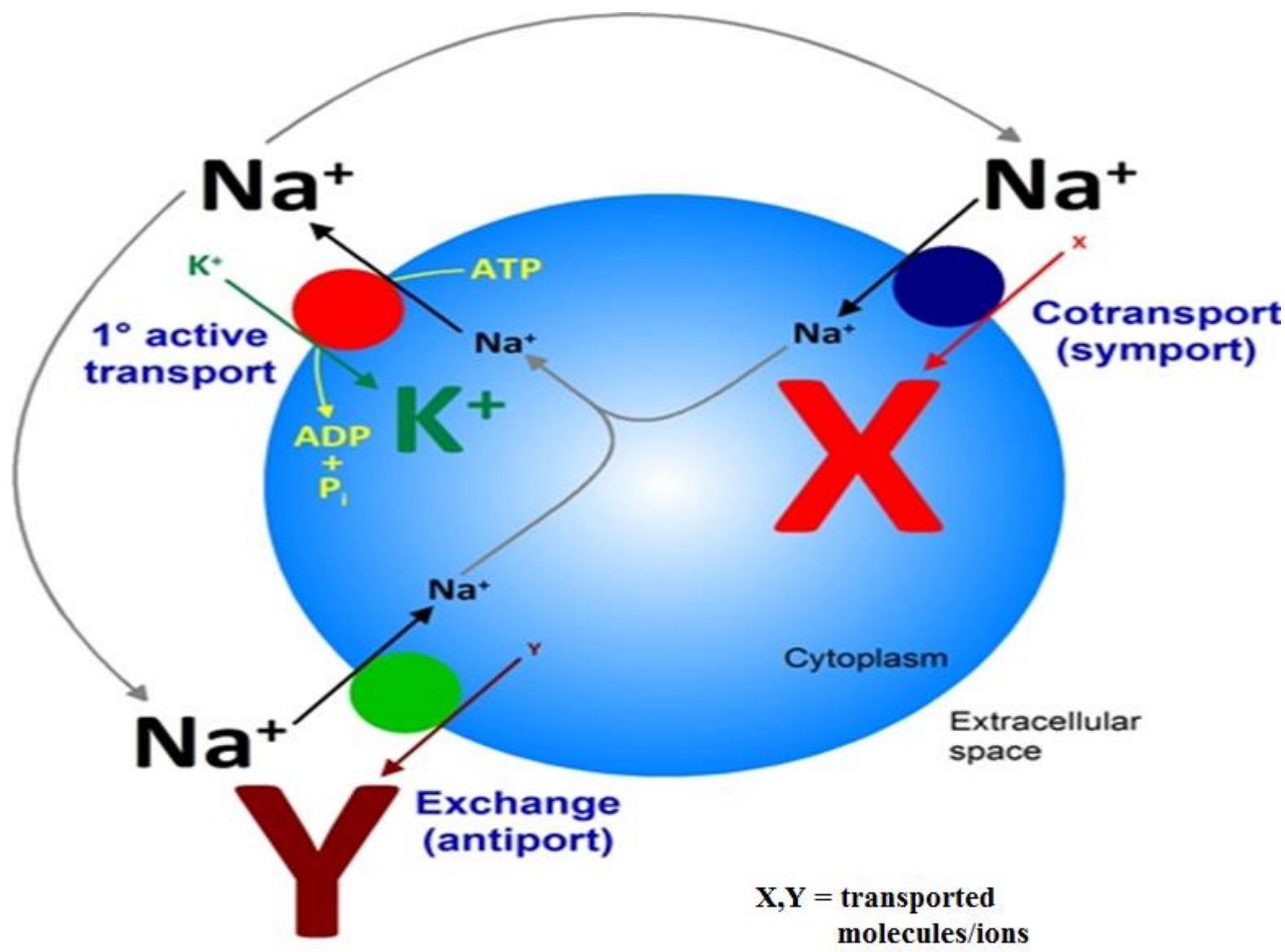
- In gastric parietal cells transports H^{+} into the lumen of the stomach against its electrochemical gradient
- Proton pump inhibitors (Omeprazole. Pantoprazole...)

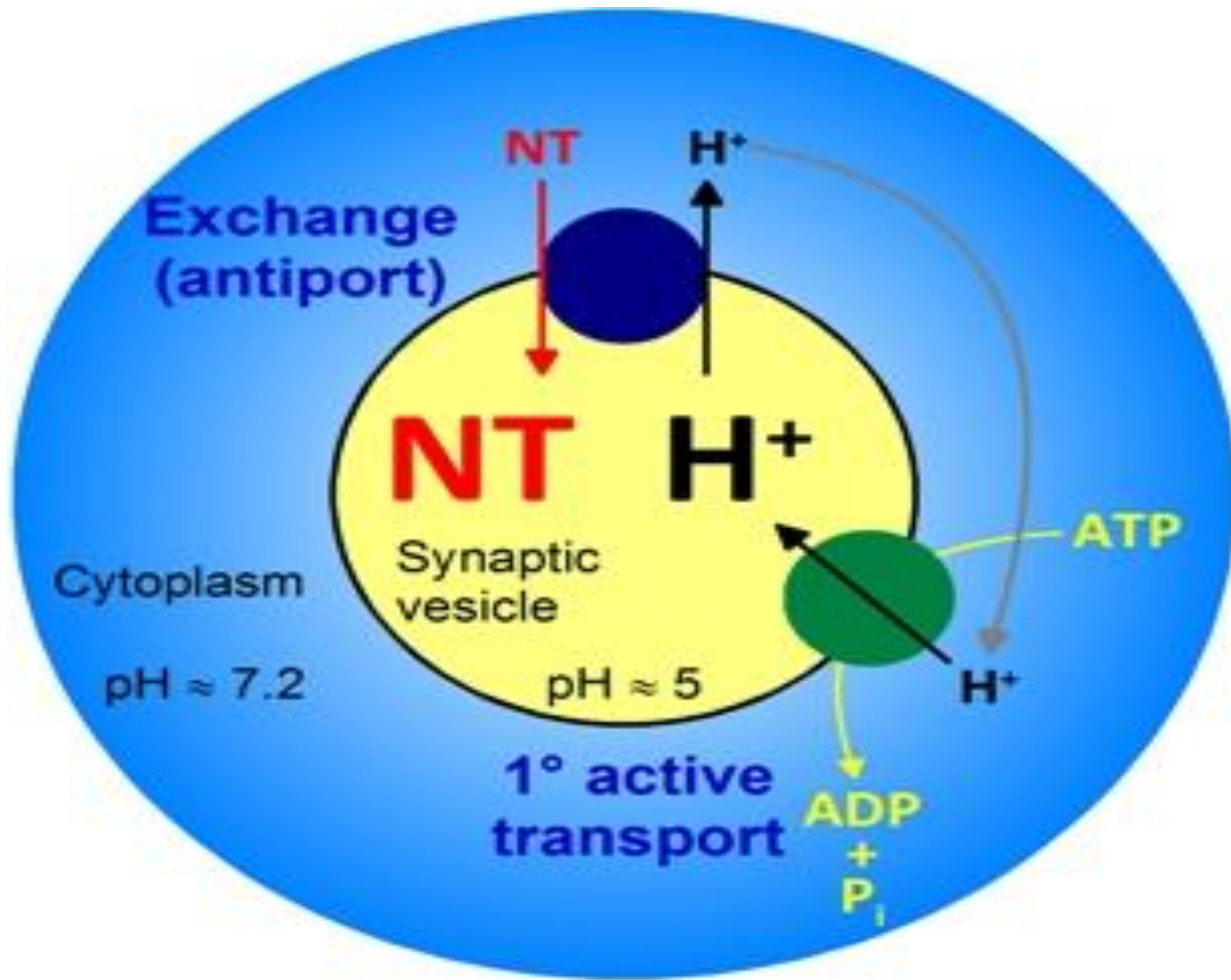
Transport Mechanisms

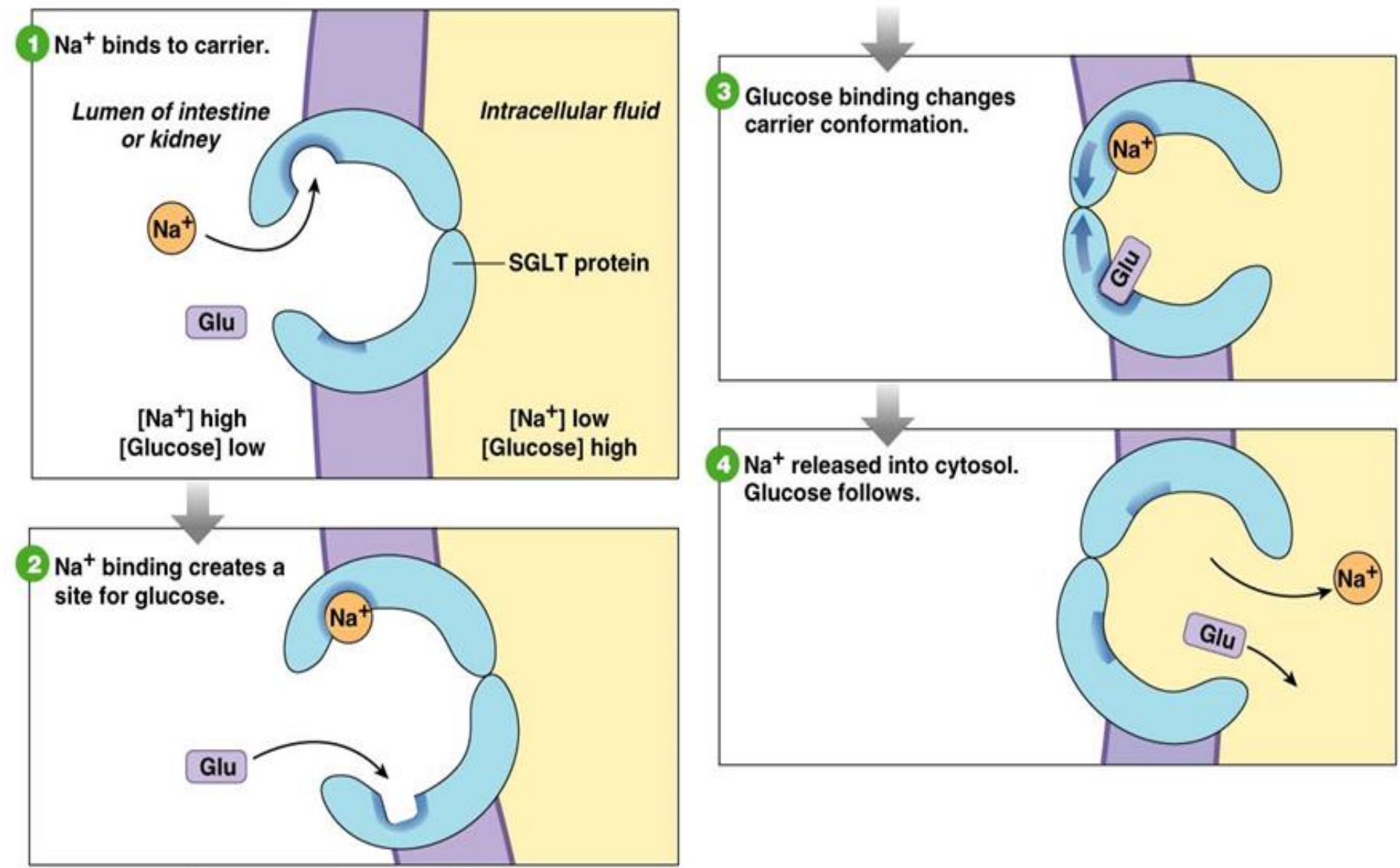
- Secondary active transport
 - A transporter protein couples the movement of an ion (typically Na^+/H^+) down its electrochemical gradient to the uphill movement of another molecule/ion
 - Uses potential energy, which is derived through exploitation of an electrochemical gradient.
 - The energy created from one ion moving down its electrochemical gradient is used to power the transport of another ion moving against its electrochemical gradient

Transport Mechanisms

- Secondary active transport: types
 - Cotransport/Symport
 - Na^+ /glucose cotransport
 - $\text{Na}^+ - \text{K}^+ - 2\text{Cl}^-$ cotransport
 - Exchange/antiport
 - $\text{Na}^+/\text{Ca}^{2+}$ exchanger
 - Na^+/H^+ counter-transport
 - H^+ /oligopeptide transporter
 - H^+/NT exchanger







Transport Mechanisms

- Secondary active transport
- **NB**
 - The energy for 2° active transport is derived from the "downhill" movement of the drive ion (Na⁺/H⁺)
 - The electrochemical gradient of the drive ions is maintained by 1° active transport
 - Dysfunction of the 1° active transporter secondarily affects the 2° active transport

Transport Mechanisms

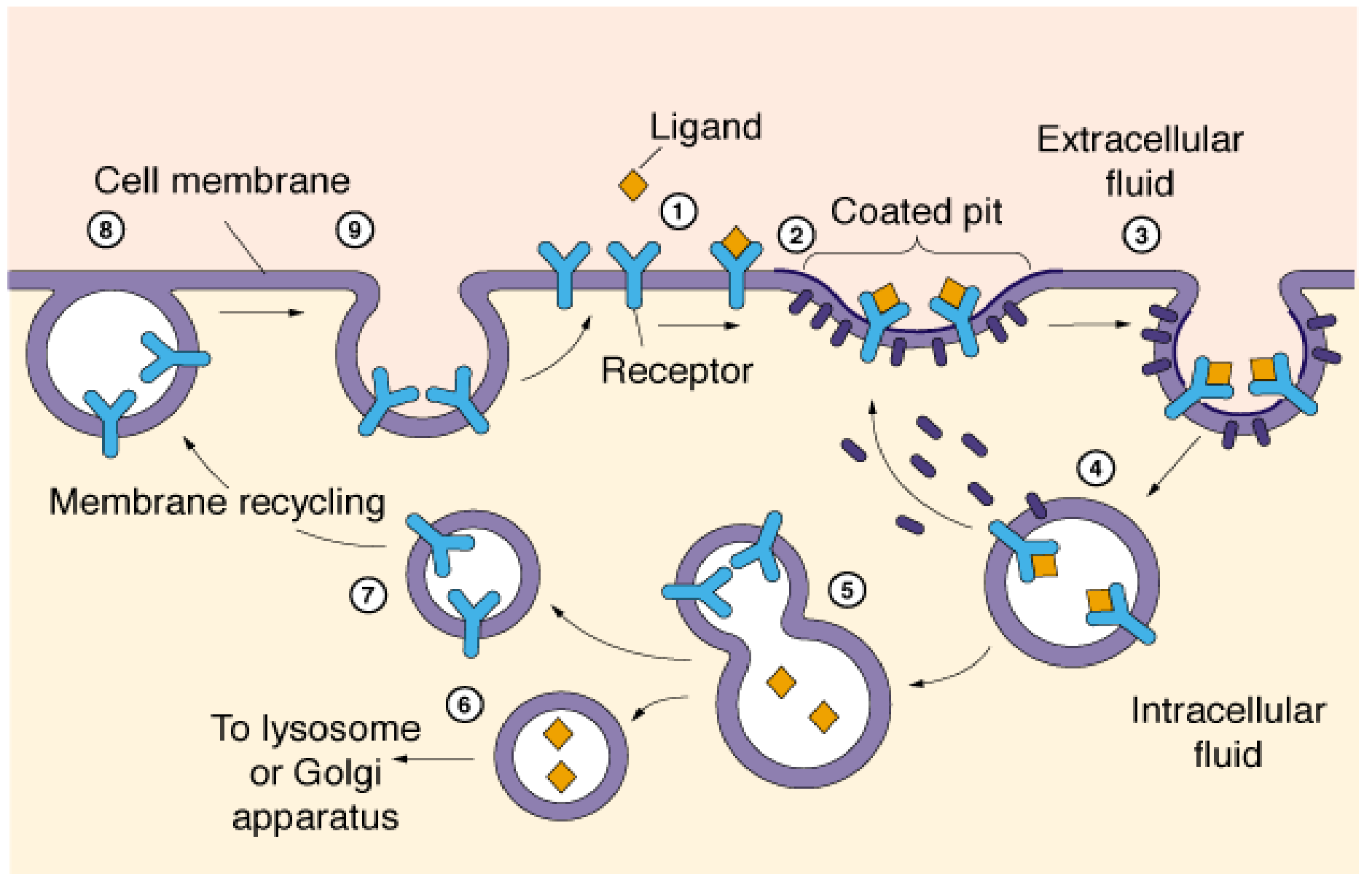
3. Bulk transport

- Transport of larger molecules
- Uses metabolic energy
- Endocytosis + Exocytosis

Transport Mechanisms

a. Endocytosis

- Engulfing of materials by invaginating the outer part of a cell membrane until it buds off within the cytoplasm
- Requires energy
- It can be:
 - **Phagocytosis** → “cell eating ”
 - Large molecules such as bacteria, dead RBC... surrounded by cell membrane and taken up
 - **Pinocytosis**



Transport Mechanisms

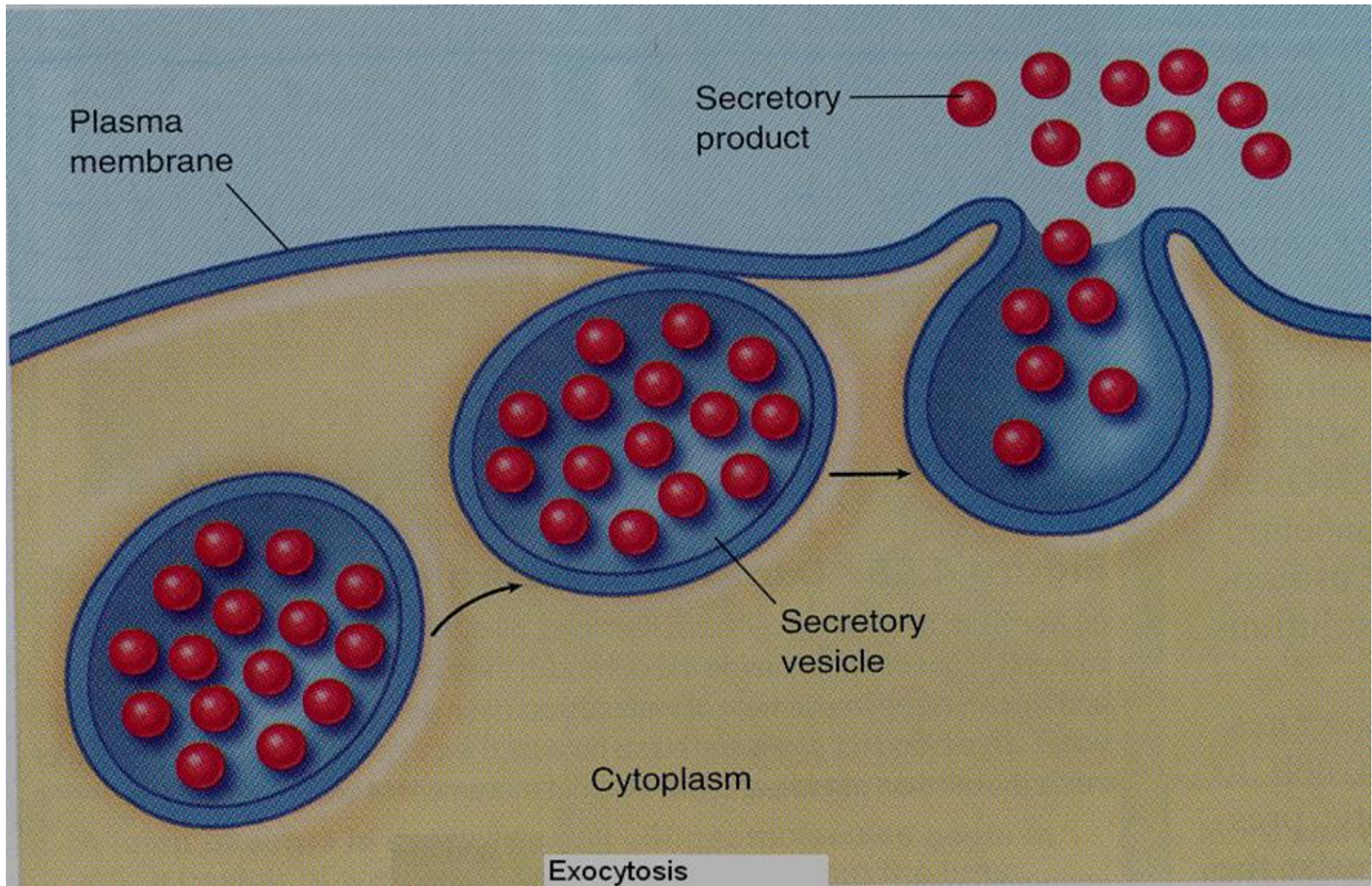
a. Endocytosis

- **Pinocytosis** → invagination occurs into cell and pinches off to form boundary of an intracellular vesicle, vacuole or tubule
- Pinocytosis may be:
 - **Fluid endocytosis** → enclosing a small volume of ECF or
 - **Adsorptive endocytosis** → specific molecules bound to cell membranes and carried into cell

Transport Mechanisms

b. Exocytosis

- **Emeiocytosis** → “Cell vomiting”
- Releases of neurotransmitters, digestive enzymes and some hormones...



Body Fluid Composition & Electrolyte Balance

Body Fluid Composition

- The plasma membrane divides the cell into compartments with specific biochemical compositions and functions
- Almost two-thirds of the body ($\approx 60\%$) consists of water
 - The remaining content is fat + fat-free solute
- Total body water is distributed in the intracellular (ICF) and extracellular compartments (ECF)
 - Wettest tissues ($\approx 70-85\%$) $\rightarrow$ muscle and grey matter of brain
 - Driest non-adipose tissues ($\approx 60\%$) $\rightarrow$ connective tissue & RBCs
 - Adipose tissue contains 20% water

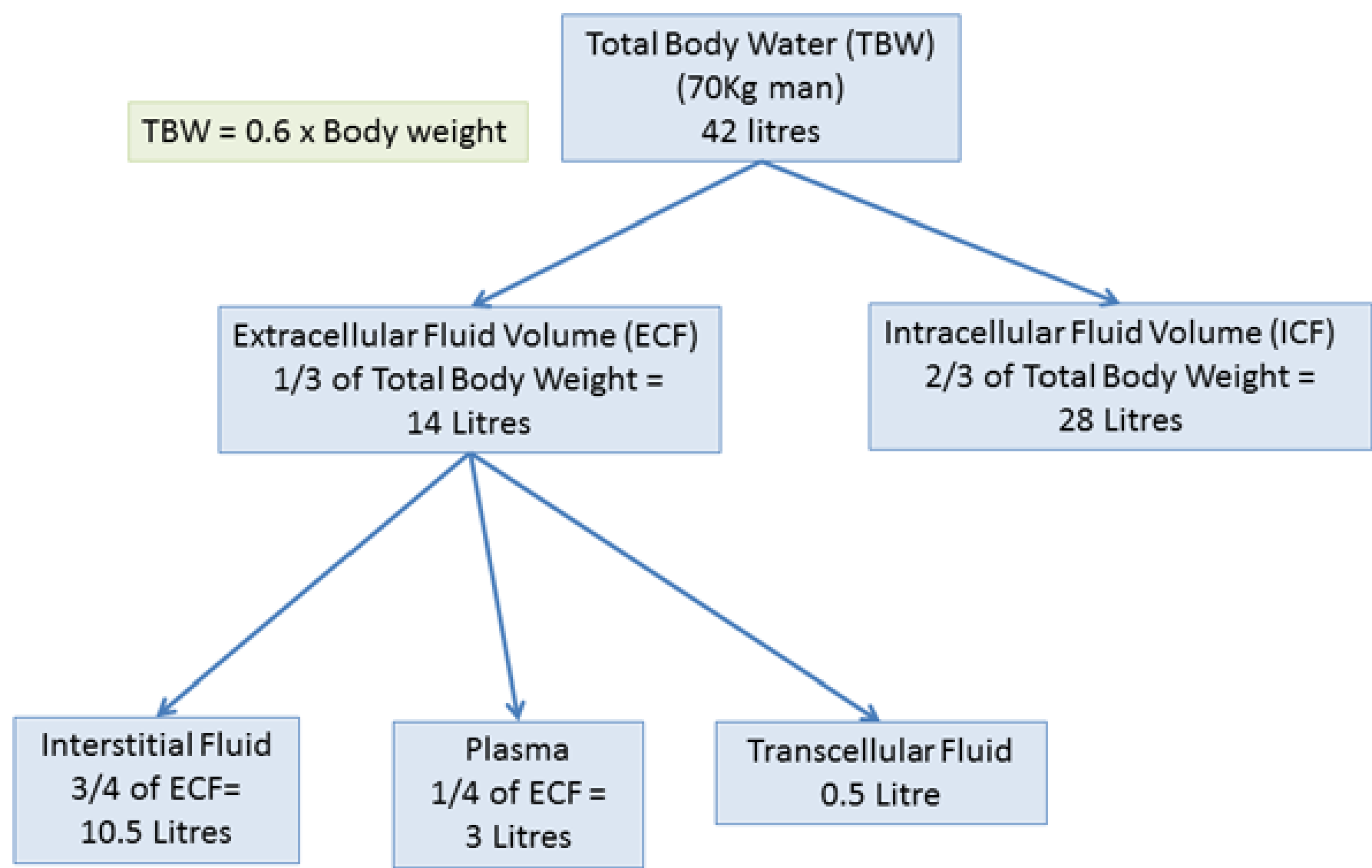
Body Fluid Composition

1. Extra-cellular Fluid (ECF)

- Constitutes about 20% of body weight (14L) $\approx$ 1/3 of TBW
 - ECF consists of **blood, lymph and transcellular fluid (CSF + fluid in eyes + ...)**
- Fluid in the vascular system $\approx$ 25% of ECF
- Fluid in the interstitium $\approx$ 75% of ECF
- Fluid in the transcellular space $\approx$ 0.5 litre

2. Intra-cellular Fluid (ICF)

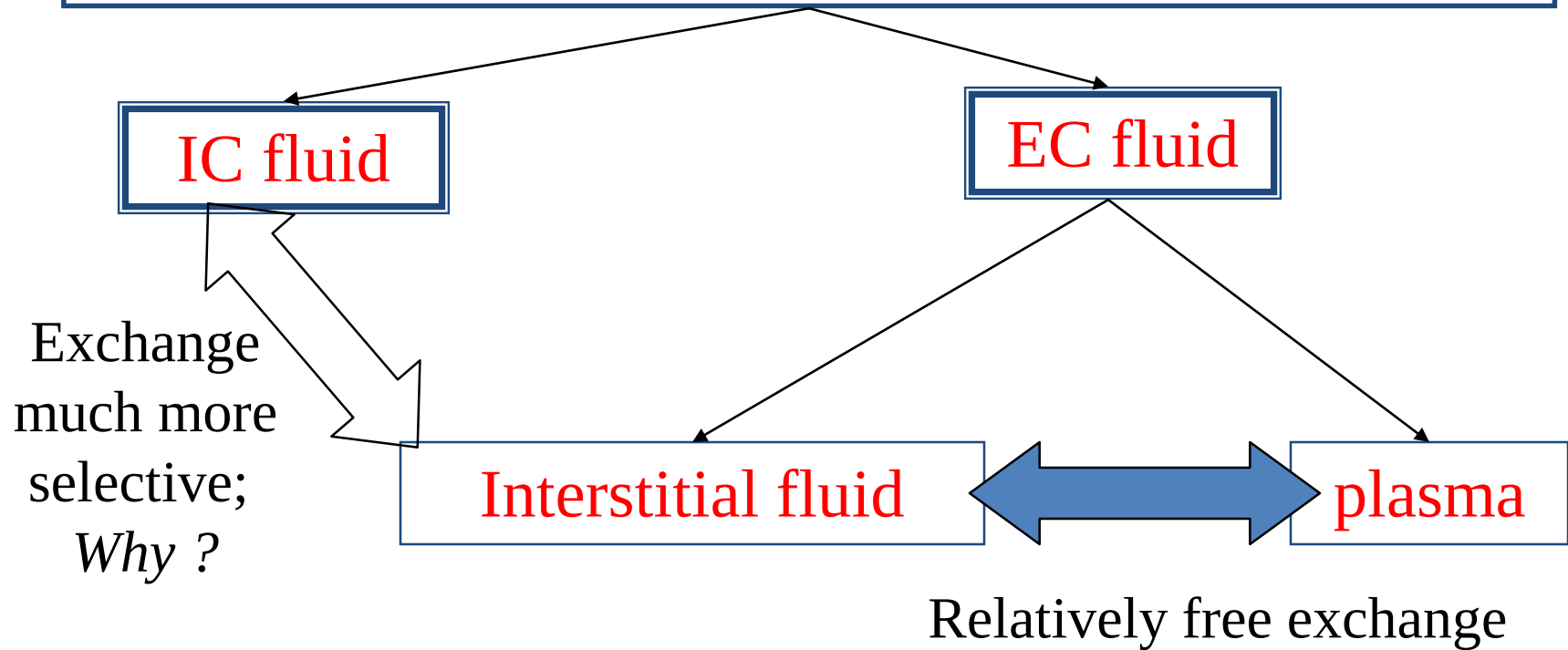
- $\approx$ 40% of body weight (28L) $\approx$ 2/3 of TBW



Body Fluid Composition

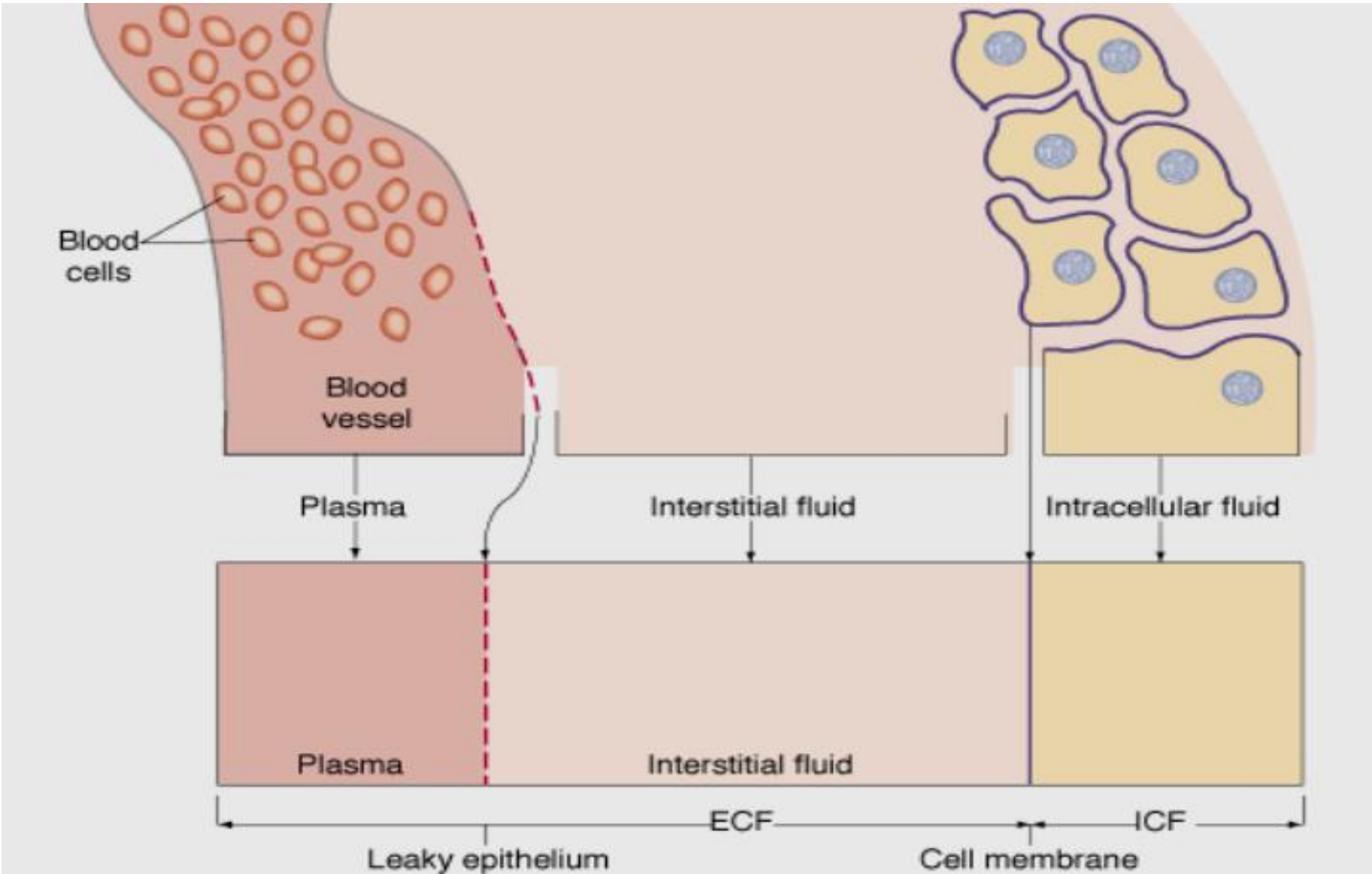
- Body Fluid Compartments
 - Each fluid compartment has the optimal biochemical composition to enable a different set of physiological processes to occur
 - Processes tend to alter composition within the compartments
 - For normal survival to continue, alterations beyond the normal range are resisted by homeostasis:
 - i.e., transport processes operate continuously to adjust and maintain constancy of the internal environment

Body Fluid Compartments



Routes of fluid transfer

- Entry point to ECF → Mouth and GIT
- Exit routes from ECF → Skin, lung, intestine and kidney



Body Fluid Compartments

Body Fluid Composition

- Body Fluid Compartments
 - Fluid compartment allows localization of particular biochemical processes in various organelles
 - Both compartments have their fluid volume and composition essentially kept constant for the cell to plod on undisturbed
 - Many cellular processes take place in or on the membranes of organelles
 - Classical examples of processes occurring on membranes of organelles are electron transport and oxidative phosphorylation

Body Fluid Composition

- Ion distribution
- Cations:
 - Major ion in ECF = Na^+
 - Major ion in ICF = K^+
 - Na-K pump helps to:
 - Push out Na^+ that leaks into cells
 - Push in K^+ that leaks out of cells

Body Fluid Composition

- Ion distribution
- Anions:
 - Large organic molecules (Proteins, phosphates) remain inside the cell (*they are too large to diffuse out*)
 - Main ECF anions → Cl^- , HCO_3^-
 - Cl^- is not exclusively distributed outside the cell → distributes according to electrical forces
 - Since many cellular anions cannot diffuse out, Cl^- tends to stay mainly outside the cells (*the Donnan effect*)

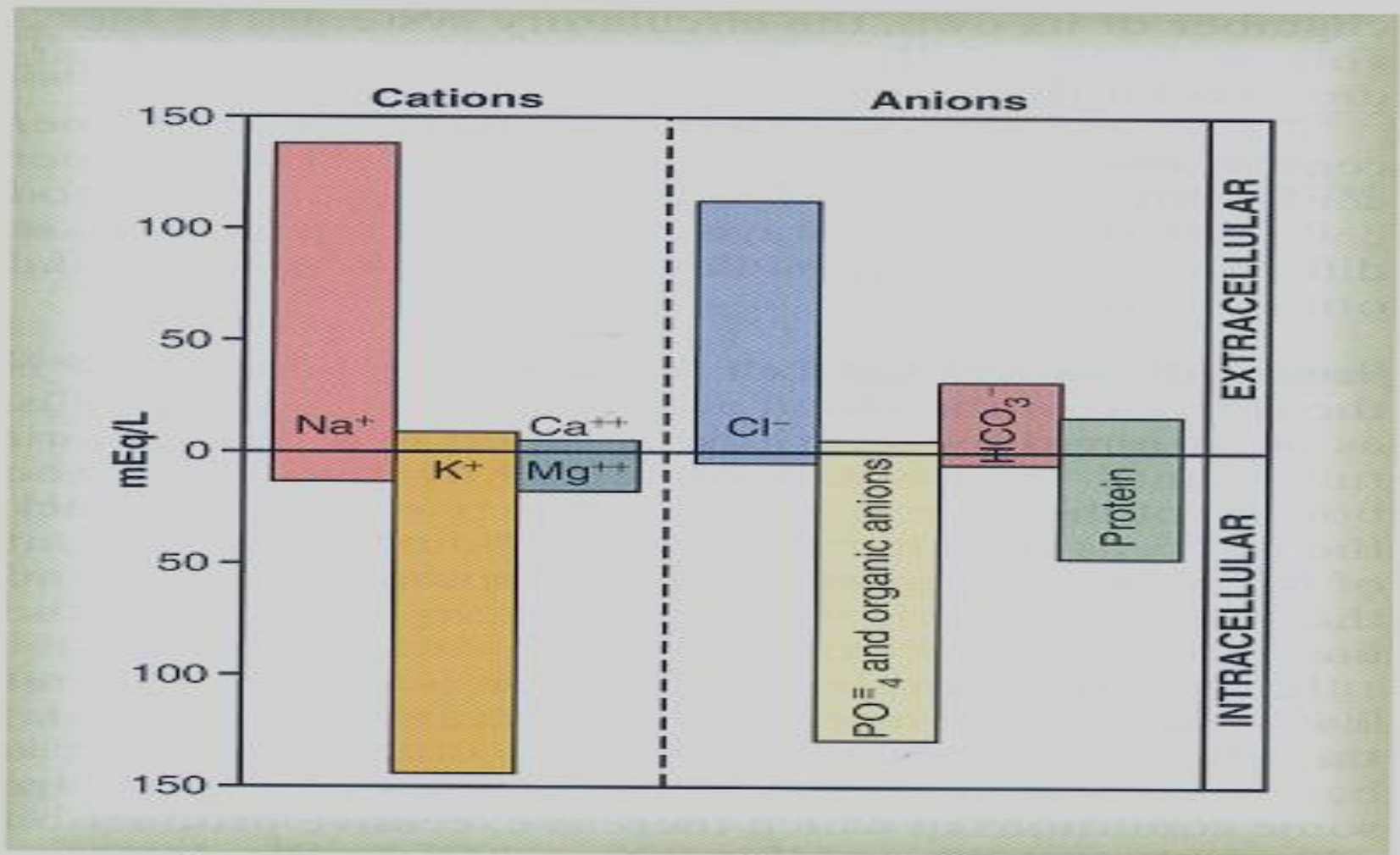
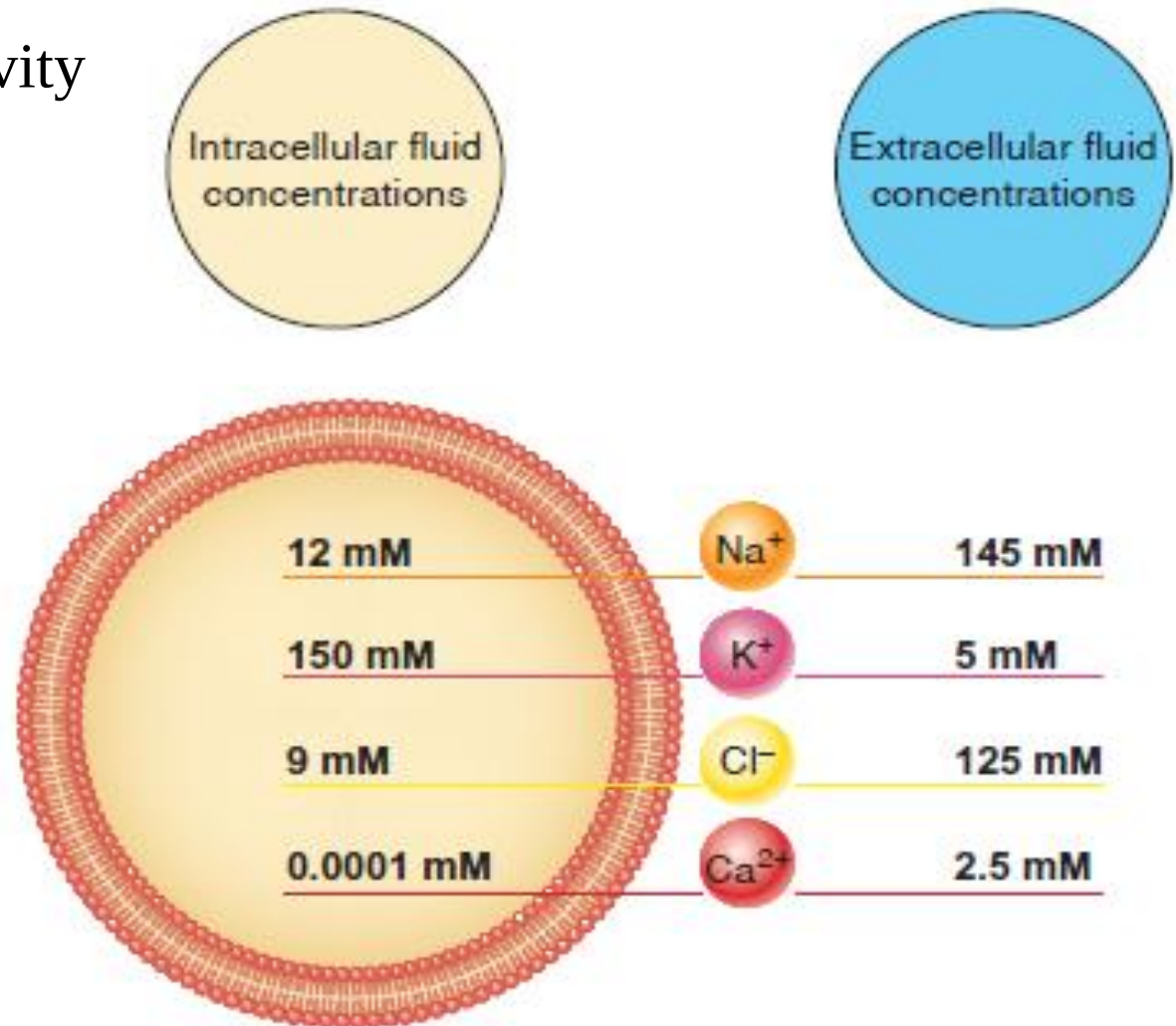


Fig..Major cations & anions of ECF &ICF

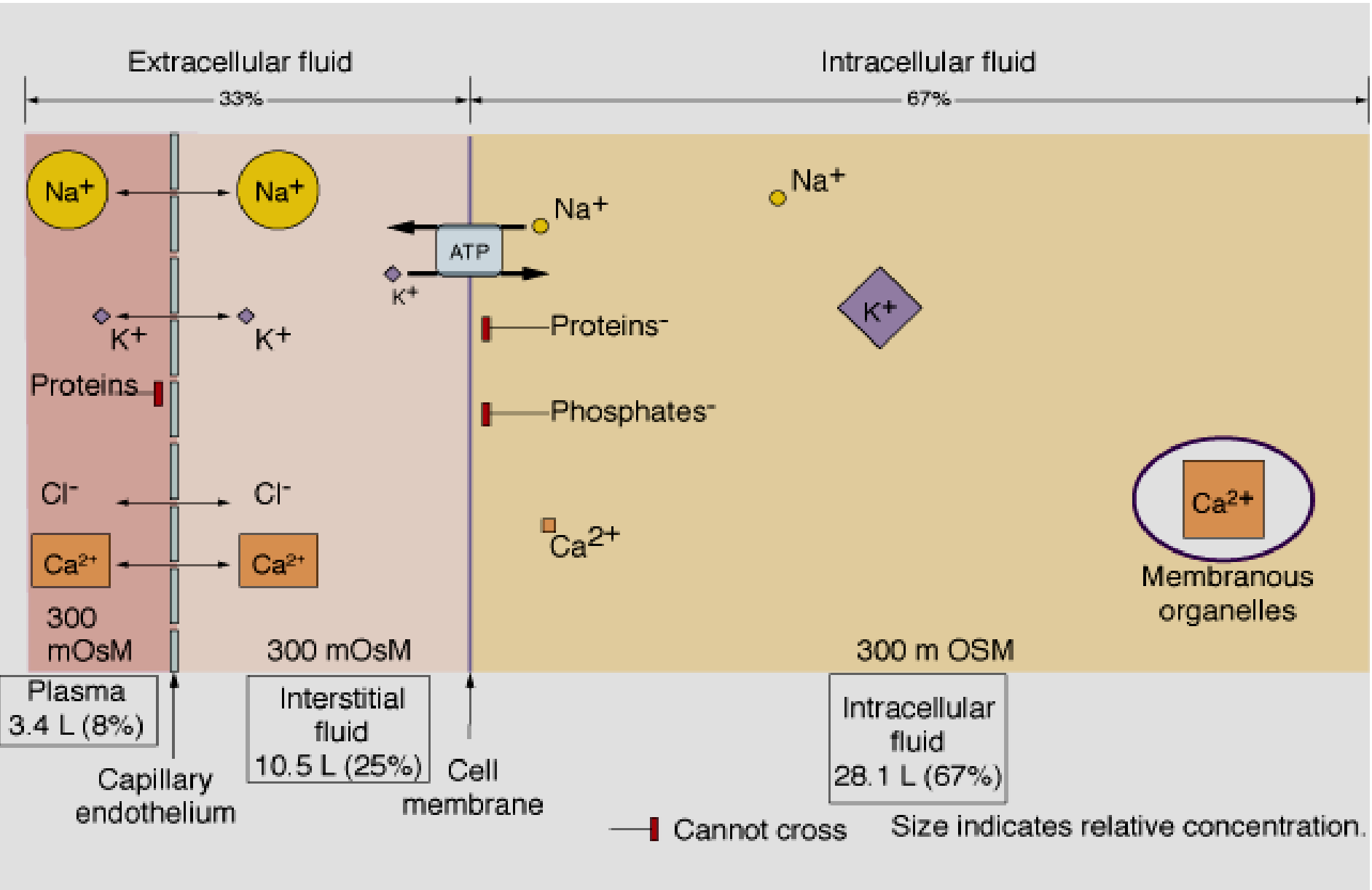
Distribution of Solutes is based on:

- Electrochemical activity
- Osmotic activity



Body Fluid Composition

- Ion distribution
 - Body fluids contain equal amounts of cations and anions because (positive charges = negative charges)
 - Distribution of ions leads to the development of:
 - Electrical activity on the cell surface
 - Osmotic pressure in cell compartments



Clinical Correlates

Edema

- **Non-pitting (cellular) edema** → fluid accumulation in the ICF
- **Pitting edema** → fluid accumulation in the IF (*firm pressure on swollen area leaves an impression*)
 - Specific edema → occurring in dependent parts
 - **Puffy face** (eyes) in the morning
 - **Puffy feet**
 - **Sacral edema** → fluid accumulation in the lower back after being in bed for several hours
 - **Anasarca** → Generalized Edema → Expanded IF volume + ↑Total body sodium

Edema

- Causes of pitting edema

1. Increased **hydrostatic pressure** in capillaries secondary to heart failure...
2. Decreased **osmotic pressure** due to low plasma proteins secondary to conditions like nephrotic syndrome
3. Increased **capillary permeability**
4. **Lymphatic obstruction**
5. Localized increase in **venous pressure** $\rightarrow$ $\uparrow$ pressure in venules, capillaries and arterioles $\rightarrow$ $\uparrow$ hydrostatic pressure $\rightarrow$ $\uparrow$ filtration

Edema

- Causes of non-pitting edema
 - Depression of cellular metabolism $\rightarrow \downarrow \text{Na}^+-\text{K}^+$ pump $\rightarrow \uparrow \text{Na}^+$ accumulation in the cell $\rightarrow$ Water following Na^+ $\rightarrow$ Cellular edema
 - Inflammation $\rightarrow \uparrow$ membrane permeability to Na^+ and other ions that diffuse into cell with water $\rightarrow$ edema
 - Water intoxication occurs when excessive volumes of water are absorbed too quickly leading to nausea, vomiting and shock-conditions
 - These are caused by undue drop in plasma osmolality before adequate inhibition of ADH secretion occurs

Edema

- **NB**

- **Depletion of ECF volume:** low plasma vol. → low BP, rapid pulse; low IF vol. → poor skin turgor, dry tongue, sunken eyes
- **Increased ECF volume** → ↑BP → pressure diuresis → excretion of Na^+ and H_2O → ↓ECF
- **Depletion of ICF vol.:** Brain goes mad with thirst due to intense stimulation of hypothalamic osmoreceptors
- **Increased ICF volume** → Water intoxication secondary to accumulation of water inside cells